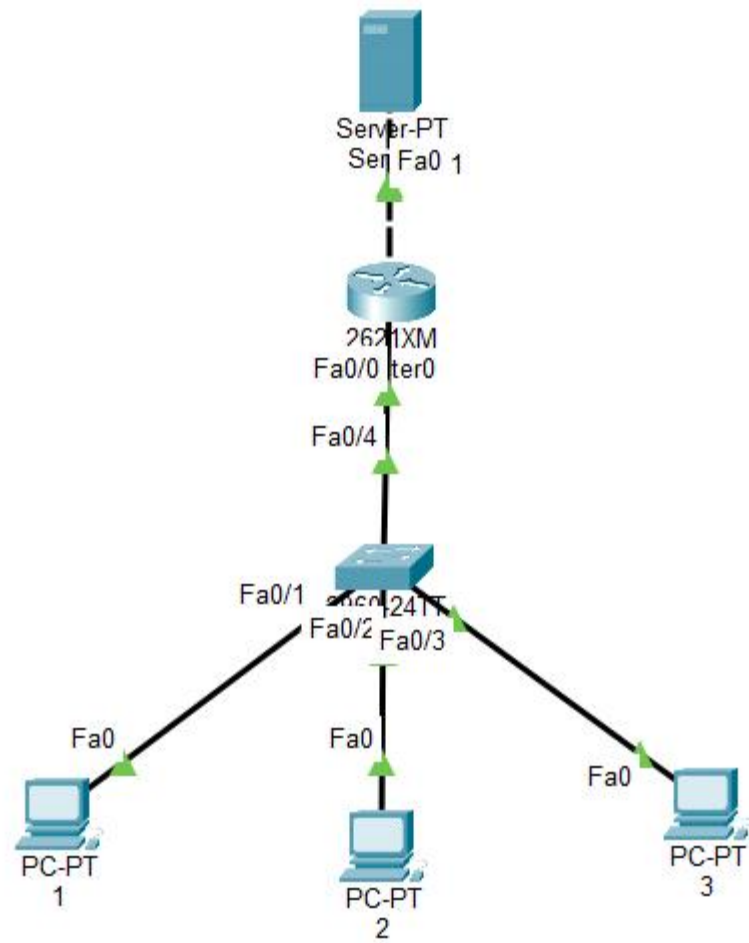
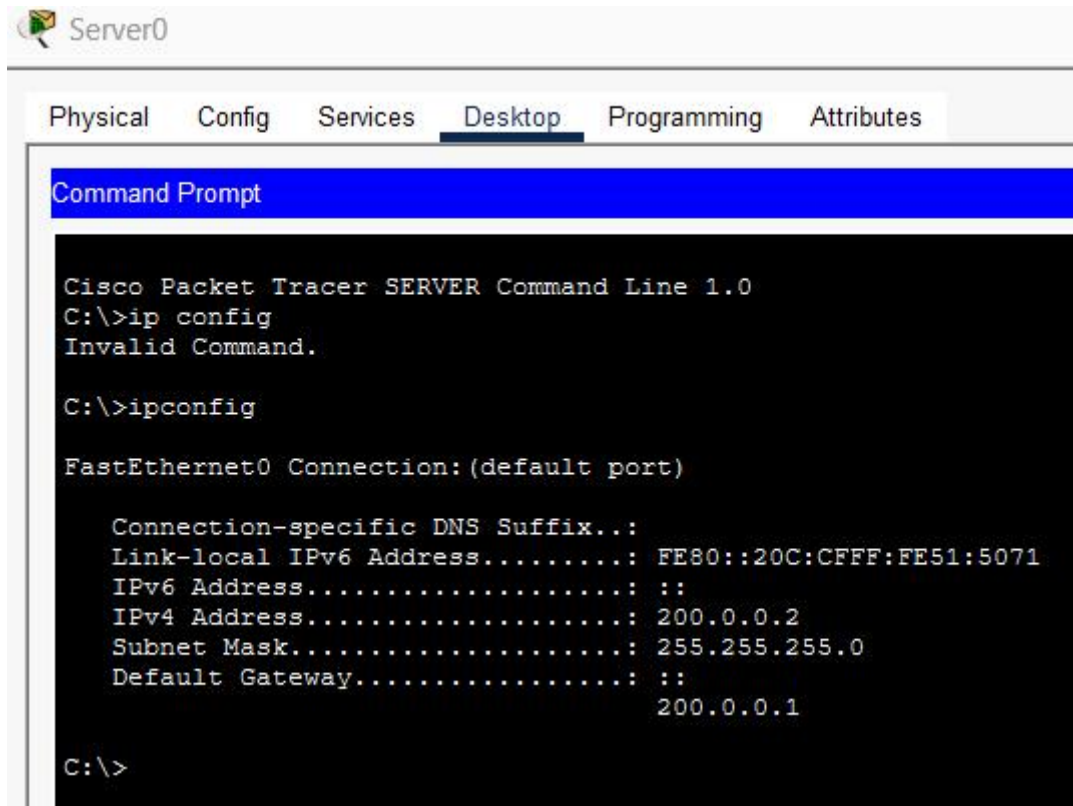


PBL 3번
구성도



서버의 ip



라우터 ip:

Router#sh ip int brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.10.1	YES	manual	up	up
FastEthernet0/1	200.0.0.1	YES	manual	up	up

Router#sh access-lists

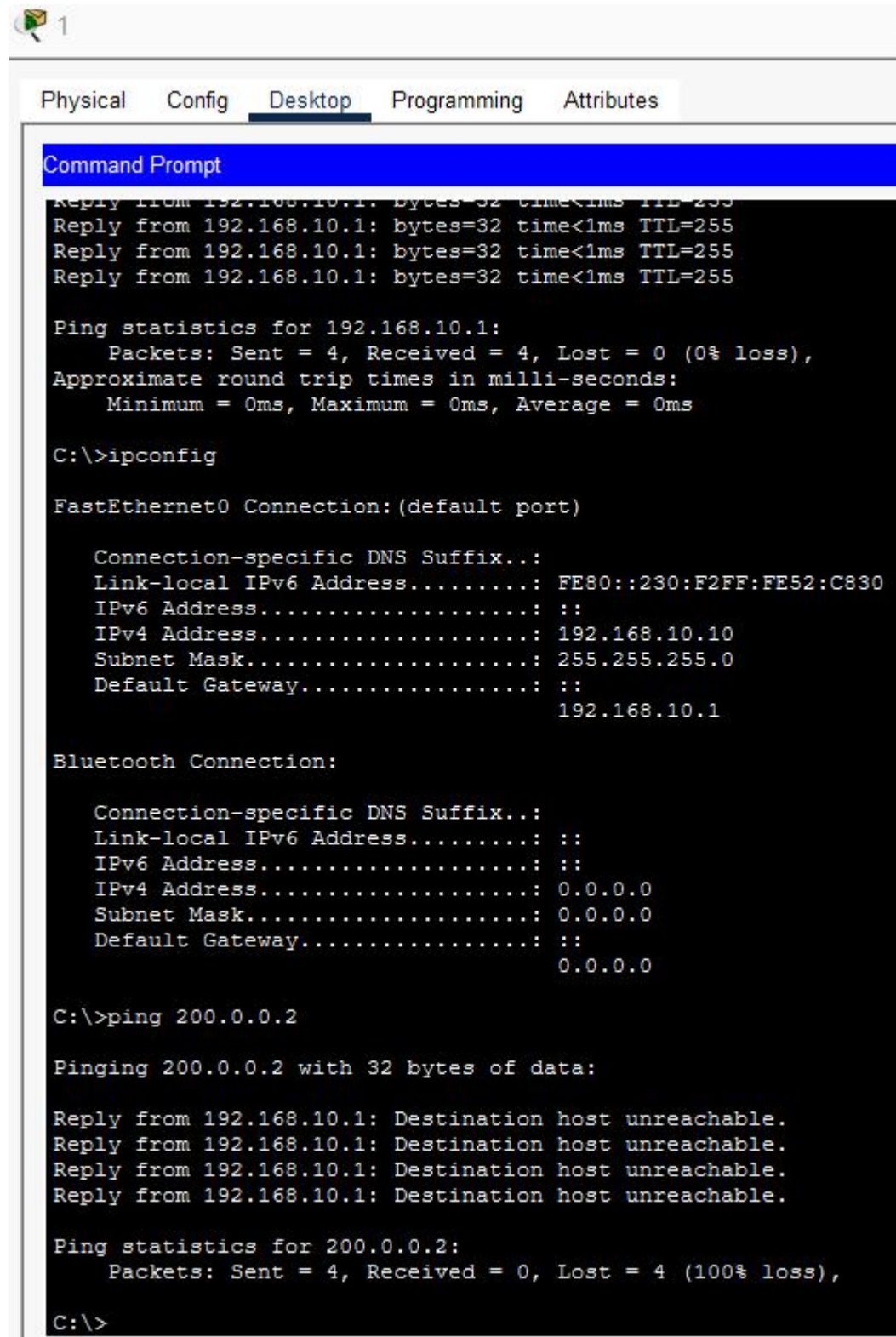
Extended IP access list 100

```
10 permit tcp 192.168.10.0 0.0.0.255 host 200.0.0.2 eq www (5 match(es))
20 permit tcp 192.168.10.0 0.0.0.255 host 200.0.0.2 eq 443
30 deny ip any any (12 match(es))
```

Router#

PC ip와 서버 핑 여부.

PC 1:



```
Command Prompt

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::230:F2FF:FE52:C830
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.10
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.10.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 200.0.0.2

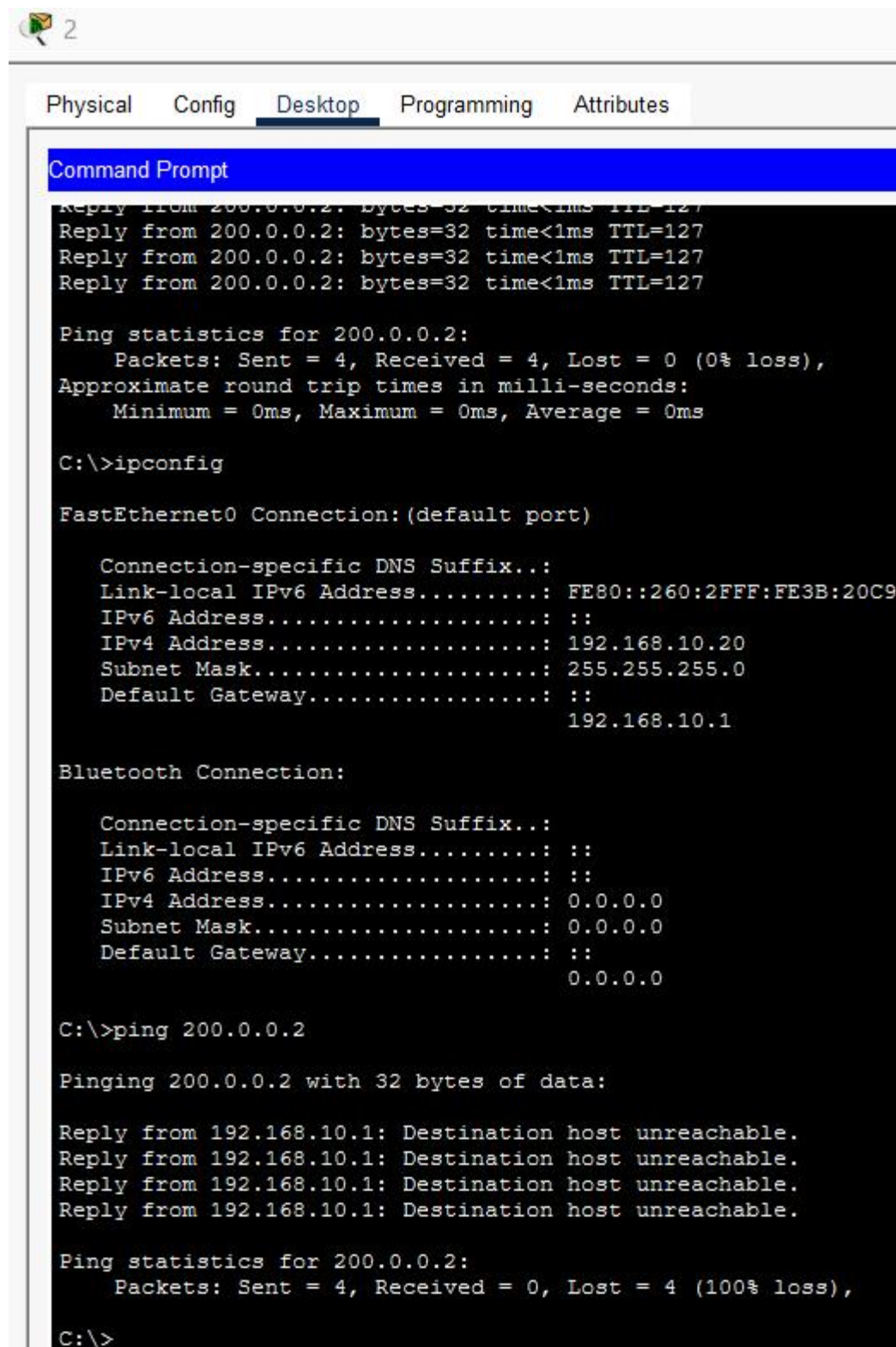
Pinging 200.0.0.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC 2:



```
2

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Reply from 200.0.0.2: bytes=32 time<1ms TTL=127
Reply from 200.0.0.2: bytes=32 time<1ms TTL=127
Reply from 200.0.0.2: bytes=32 time<1ms TTL=127
Reply from 200.0.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:2FFF:FE3B:20C9
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.20
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.10.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 200.0.0.2

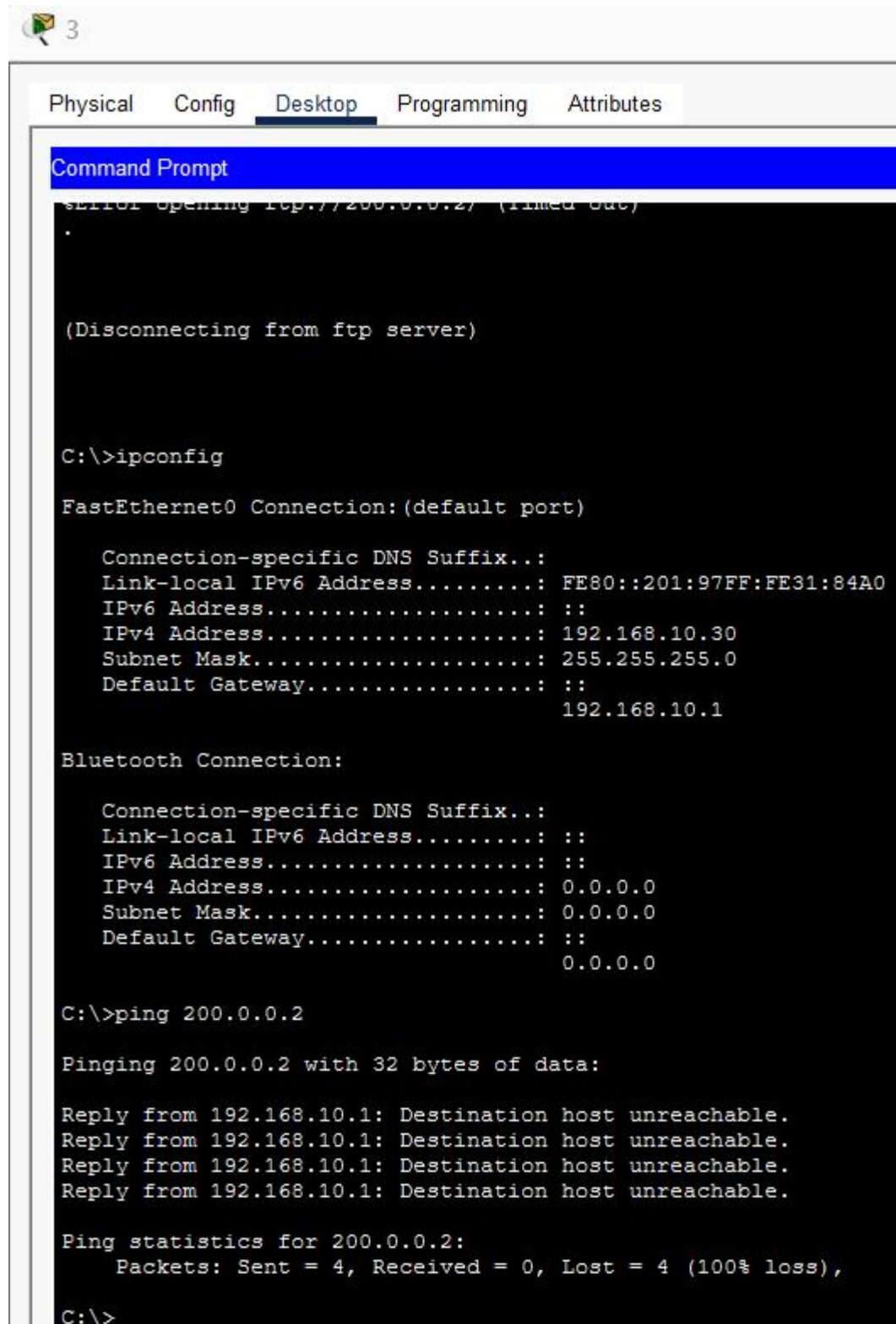
Pinging 200.0.0.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC 3:



```
3
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Error Opening Tcp://200.0.0.2/ (Timed Out)
.

(Disconnecting from ftp server)

C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: FE80::201:97FF:FE31:84A0
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.10.30
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                           192.168.10.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                           0.0.0.0

C:\>ping 200.0.0.2

Pinging 200.0.0.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC > 서버 telnet 결과

```
C:\>telnet 200.0.0.2 80
Trying 200.0.0.2 ...Open

[Connection to 200.0.0.2 closed by foreign host]
C:\>telnet 200.0.0.2 443
Trying 200.0.0.2 ...Open

[Connection to 200.0.0.2 closed by foreign host]
C:\>telnet 200.0.0.2 21
Trying 200.0.0.2 ...
% Connection timed out; remote host not responding
C:\>telnet 200.0.0.2 81
Trying 200.0.0.2 ...
% Connection timed out; remote host not responding
C:\>
```

PC 웹 브라우저 접속도 잘 되는 것을 확인할 수 있었다.



보고서: 내부망 사용자가 웹 서비스만 이용할 수 있도록 방화벽 ACL을 설정하였다.
HTTP(80), HTTPS(443) 허용.